Representation of Perceptual Uncertainty in Mouse V1

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Methods

Visual Stimulus Generation

Visual stimuli consisted of static, full-screen square-wave gratings generated programmatically in MATLAB. To maintain constant visual angles across the visual field, spatial frequencies were calculated based on the physical dimensions of the monitor (53.2 cm width) and the animal's viewing distance (20 cm), yielding a spatial frequency of 0.04 cycles per degree.

To manipulate sensory uncertainty, stimuli varied along three dimensions: base orientation (θ), spatial dispersion (d), and contrast (c). Base orientations were drawn from a 9-angle set covering the first quadrant ($0°, 15°, 30°, 40°, 45°, 50°, 60°, 75°, 90°$). To construct stimuli with varying levels of dispersion, 9 component sine-wave gratings were generated over a 360° space centered on the base orientation. The amplitude of each component was weighted according to a Gaussian distribution, where the standard deviation equaled the designated dispersion parameter ($d \in \{5°, 30°, 45°, 90°\}$). The weighted components were summed, and the resulting composite image was thresholded using a signum function to produce a sharp, square-wave edge profile. Finally, the amplitude of the composite grating was scaled to achieve the target contrast level (1%, 25%, 50%, 100%) around a 0.5 mean luminance gray background.

Session Structure and Stimulus Sampling

Mice were trained in progressive stages, beginning with a "Basic" task utilizing only the easiest, orthogonal stimuli (0° vs 90° at maximum 100% contrast and minimum 5° dispersion). Following basic task learning, animals progressed through intermediate training stages: first generalizing across orientation only (initially introducing 15° and 75°, then 30° and 60°, and finally the near-boundary 40°, 45°, and 50° orientations). Subsequently, mice were exposed to sessions manipulating individual stimulus dimensions in discrete stages (first contrast manipulations, then dispersion manipulations). Finally, mice advanced to "Full" sessions designed to probe the entire parametric space concurrently.

During "Full" sessions, trial sequences were dynamically generated to balance extensive parametric sampling with sustained animal motivation. Each session commenced with a brief warm-up block (e.g., 50 trials) of unambiguous, high-contrast orthogonal orientations. Following this block, the algorithm sampled from the complete grid of orientations and contrast-dispersion pairs. To prevent behavioral disengagement caused by extended periods of high-uncertainty stimuli, contrast and dispersion pairs were sampled using a geometrically weighted probability distribution (scalar factor = 0.5). This biased the presentation toward easier stimuli (lower dispersion, higher contrast) while still probing the highly uncertain regimes.

To prevent the development of stereotypic spatial or licking biases, pseudo-randomization enforced a strict trial sequence constraint: the animal experienced a maximum run length of 7 consecutive rewarded or non-rewarded trials. The decision boundary was fixed at 45°, and trials presented exactly at this boundary were ambiguously assigned to the Go or No-Go category with a 50% probability on a trial-by-trial basis.

Behavioural Task Logic and Confidence Probing

Each trial was initiated with a 2-second open-loop presentation of the static grating stimulus. Following this sensory epoch, the animal entered a closed-loop virtual corridor rendered via the ViRMEn engine. Locomotion on the freely rotating wheel was read via an optical encoder, advancing the animal's virtual position.

The corridor contained a predefined reward zone spanning 100 to 140 virtual units. On trials belonging to the 'Go' category, the animal was required to lick a capacitive spout while inside the reward zone to trigger a water reward. To behaviourally assay the animal's confidence, the exact number of licks required to elicit the reward was drawn from a geometric distribution. This required the animal to exhibit sustained, confident licking to receive the water drop, preventing simple impulsive licking strategies.

Conversely, on 'No-Go' trials, any lick detected within the reward zone constituted a False Alarm, which was followed by a 4-second timeout (rendered as a completely dark virtual environment) once the animal reached the end of the corridor. The current trial concluded when the animal reached a randomly drawn end-of-corridor position between 300 and 350 virtual units. At this point, the animal was seamlessly teleported to the start of the next trial (in the case of a Correct Rejection or a completed Go trial) or entered the 4-second timeout period (in the case of a False Alarm) before the new trial could begin.

Calcium Imaging and Spike Deconvolution

We performed two-photon calcium imaging of Layer II/III V1 population activity expressing the genetically encoded calcium indicator GCaMP8. Because raw calcium fluorescence represents a low-pass filtered proxy of the underlying action potentials, fluorescence traces were computationally deconvolved into inferred spike rates. We utilised CASCADE, a supervised deep learning-based deconvolution algorithm [1]. Specifically, we applied a CASCADE model recently optimised and calibrated for the specific binding kinetics and signal-to-noise characteristics of GCaMP8 indicators [2]. This procedure allowed us to extract robust, high-fidelity estimates of instantaneous spike probabilities, which subsequently served as the input state vectors ($\mathbf{r}_{n,t}$) for our temporal decoding architectures.

Longitudinal ROI Registration

To investigate the stability of neural representations across varying stimulus conditions (e.g., contrast and dispersion) and task generalisation stages, we tracked the same populations of V1 neurons across multiple recording sessions. For each longitudinally tracked animal, we designated a single reference session (typically the earliest baseline session in a given sequence). We then systematically performed spatial alignment and pairwise Region of Interest (ROI) matching between this reference session and all subsequent target sessions.

Specifically, pairwise mappings of cell indices between the reference field-of-view and each target field-of-view were computed using a semi-automatic, user-led ROI matching pipeline (custom open-source software available at `https://github.com/rochefort-lab/roi_matching`). For each session pair, the process was initialized by the user manually identifying a small set of high-confidence corresponding neurons to serve as spatial seeds. The algorithm subsequently utilised these user-provided seeds to estimate the optimal global spatial transformation (accounting for translation and rotation) between the two imaging planes. Following this alignment, the algorithm automatically resolved the remaining matches across the wider population by evaluating spatial proximity and nearest-neighbour anatomical heuristics.

To ensure a highly stable neural population for across-day analysis, we calculated the strict intersection of these pairwise matches. Only neurons that were successfully and unambiguously matched across *all* evaluated sessions for a given animal were retained. This rigorous intersection criterion yielded a conservative set of common neuron identities, allowing us to evaluate population state vectors within a fixed-dimensional neural subspace across days without the confounding influence of neural dropout or false-positive matching.

Ideal Observer Model and Behavioural Fitting

To estimate the animal's trial-by-trial perceptual posterior, we formulated an Ideal Observer (IO) model that assumes the animal makes a noisy internal measurement, m, of the true stimulus orientation, s, and subsequently calculates a decision variable to drive behaviour.

Generative Model and Inference

We modelled the internal measurement m as being drawn from a von Mises distribution centred on the true stimulus orientation:

$$p(m \mid s) = \text{VonMises}(m; s, \kappa(c, d)) = \frac{\exp(\kappa \cos(2(m - s)))}{2\pi I_0(\kappa)} \tag{1}$$

Crucially, the precision of the sensory evidence, κ, was modelled as an isotropic function of the stimulus contrast (c) and dispersion (d):

$$\kappa(c, d) = (\kappa_{min} + \kappa_{amp}) \cdot c^{p_c} \cdot \exp(-p_d \cdot d) \tag{2}$$

where κ_{amp}, p_c, and p_d are free parameters controlling the dynamic range and scaling of sensory precision.

Upon observing m, the observer computes a perceptual posterior over the stimulus orientation using Bayes' theorem:

$$p(s \mid m) = \frac{p(m \mid s)p(s)}{\int p(m \mid s')p(s')ds'} \tag{3}$$

where the prior $p(s)$ reflects the asymmetric task statistics, modelled as an equal-weighted bimodal von Mises mixture peaking at the 0° and 90° target orientations:

$$p(s) = \frac{1}{2}\text{VonMises}(s; 0°, \kappa_{prior}) + \frac{1}{2}\text{VonMises}(s; 90°, \kappa_{prior}) \tag{4}$$

To translate this perceptual inference into a choice, we formally define the objective task contingencies as a utility function $U(A, s)$, which maps a given stimulus orientation s and action $A \in \{\text{Go}, \text{No-Go}\}$ to a scalar reward or cost. Given the category boundary at 45°, the utility for the 'Go' action is defined as a step function:

$$U(\text{Go}, s) = \begin{cases} R_{\text{Hit}} & \text{if } s < 45° \\ 0.5 \cdot R_{\text{Hit}} + 0.5 \cdot R_{\text{FA}} & \text{if } s = 45° \\ R_{\text{FA}} & \text{if } s > 45° \end{cases} \tag{5}$$

and conversely for the 'No-Go' action:

$$U(\text{No-Go}, s) = \begin{cases} R_{\text{Miss}} & \text{if } s < 45° \\ 0.5 \cdot R_{\text{Miss}} + 0.5 \cdot R_{\text{CR}} & \text{if } s = 45° \\ R_{\text{CR}} & \text{if } s > 45° \end{cases} \tag{6}$$

where R denotes the respective empirical reward or punishment values.

The expected utility (EU) of an action, given the latent measurement m, is computed by marginalising the utility function over the perceptual posterior:

$$EU(A \mid m) = \int U(A, s)p(s \mid m)ds \tag{7}$$

A continuous Decision Variable (DV) is then formally defined as the difference in expected utility between the competing actions:

$$DV(m) = EU(\text{Go} \mid m) - EU(\text{No-Go} \mid m) \tag{8}$$

Continuous Kinematics as Confidence Proxies

Rather than fitting the Ideal Observer strictly to binary choices—which discards rich trial-to-trial variance—we leveraged continuous kinematic readouts—pre-reward zone licks (y_{lick}) and wheel velocity (y_{vel})—as empirical proxies for decision confidence. To account for baseline variations in motor vigour across days and subjects, these kinematic streams were robustly Z-scored independently for each animal, converting them into continuous, normally distributed variables.

These normalized confidence streams were modelled as conditionally independent, linear functions of the internal decision variable subject to Gaussian noise:

$$p(y_{lick} \mid m) = \mathcal{N}(y_{lick};\ \beta_{lick} DV(m) + \alpha_{lick},\ \sigma^2_{lick}) \tag{9}$$

$$p(y_{vel} \mid m) = \mathcal{N}(y_{vel};\ \beta_{vel} DV(m) + \alpha_{vel},\ \sigma^2_{vel}) \tag{10}$$

By fitting the model strictly to these continuous confidence variables, we were able to rigorously evaluate the generative power of the model by testing its ability to forecast the animal's discrete binary choices (Go/No-Go) out-of-sample, without direct parameter optimisation on the choice data itself.

Hierarchical Fitting and Marginalised Trial-by-Trial Inversion

Parameters were optimised to maximise the joint likelihood of the continuous confidence readouts using Bayesian Adaptive Direct Search (BADS). To prevent overfitting, we employed a two-stage hierarchical cross-validation procedure: first estimating a stable global parameter template strictly on the training folds pooled across all sessions from all animals, and subsequently utilising these global parameters to seed the fits for individual subjects.

Following parameter optimisation, we performed trial-by-trial model inversion to infer the ground-truth target distribution for the neural decoders, $Q(\theta)$. Crucially, rather than collapsing the observer's internal state to a single point estimate, we maintained the full uncertainty over the agent's latent internal measurement, m. Given the generative prior and the observed kinematics on trial t, the posterior probability over the internal measurement was computed as:

$$p(m \mid s_{true,t}, y_{lick,t}, y_{vel,t}) = \frac{p(y_{lick,t} \mid m) p(y_{vel,t} \mid m) p(m \mid s_{true,t})}{\int p(y_{lick,t} \mid m') p(y_{vel,t} \mid m') p(m' \mid s_{true,t}) dm'} \tag{11}$$

The idealised perceptual posterior for trial t, $Q(\theta)$, was then derived by explicitly marginalising the measurement-conditioned perceptual posteriors over this distribution of likely internal measurements:

$$Q(\theta) = \int p(s = \theta \mid m) \cdot p(m \mid s_{true,t}, y_{lick,t}, y_{vel,t}) dm \tag{12}$$

This fully marginalised distribution reflects the expected perceptual belief of the animal on that specific trial, serving as the ground-truth target for our artificial neural network decoders.

Quantification of Perceptual and Decision Uncertainty

Following the trial-by-trial inversion, we quantified both perceptual and decision uncertainty directly from the marginalised posterior, $Q(\theta)$. In accordance with normative probabilistic frameworks [3], we draw a formal computational distinction between uncertainty regarding the objective sensory stimulus and uncertainty regarding the choice-relevant categorical state.

Perceptual Uncertainty

Perceptual uncertainty captures the spread of the observer's belief regarding the exact stimulus orientation, independent of the task contingencies. We defined this as the standard deviation of the marginalised perceptual posterior, $Q(\theta)$. Because the stimulus orientation s spans a bounded domain in our task paradigm, we quantified this dispersion using the linear standard deviation:

$$U_{\text{perceptual},t} = \sqrt{\int Q(\theta) \left(\theta - \bar{\theta}\right)^2 d\theta} \tag{13}$$

where $\bar{\theta}$ is the expected value (mean) of the distribution.

Decision Posterior and Decision Uncertainty

While perceptual uncertainty reflects a task-independent sensory representation, decision uncertainty reflects the ambiguity of the choice-relevant latent state (Go vs. No-Go), which is explicitly dependent on the 45° category boundary. To quantify this task-dependent uncertainty, we first computed the marginalised categorical probability that the stimulus belongs to the 'Go' category by integrating $Q(\theta)$ over the target boundary:

$$P(\text{Go}_t) = \int_0^{45^\circ} Q(\theta)d\theta + 0.5 \cdot Q(45^\circ) \tag{14}$$

The probability of a 'No-Go' stimulus was the complement:

$$P(\text{No-Go}_t) = 1 - P(\text{Go}_t) \tag{15}$$

Finally, decision uncertainty was formally quantified as the Shannon entropy of this binary category posterior:

$$U_{\text{decision},t} = -P(\text{Go}_t)\log_2 P(\text{Go}_t) - P(\text{No-Go}_t)\log_2 P(\text{No-Go}_t) \tag{16}$$

By utilizing Shannon entropy, this formulation explicitly tracks the inverse certainty of the discrete, choice-relevant variable [3]. Decision uncertainty peaks precisely when the integrated perceptual belief is evenly split across the task boundary (maximum entropy), reflecting states where the optimal choice is highly ambiguous despite the underlying perceptual variance.

Ideal Observer Model Validation and Parameter Recovery

To validate the mathematical identifiability of our model, we conducted a rigorous parameter recovery analysis. We generated a cohort of synthetic animals parametrised both by completely random parameter assignments and by the empirically fitted parameters derived from the experimental animals. For each synthetic animal, we simulated trial-by-trial internal measurements and generated corresponding continuous confidence readouts (licks and velocity). These synthetic behavioural streams were subsequently passed through our standard hierarchical fitting and marginalised inversion pipeline blind to the generative parameters. Model fidelity was quantified by calculating the coefficient of determination (R^2) between the true and recovered values for each free parameter, alongside computing the trial-by-trial Kullback-Leibler (KL) divergence between the ground truth synthetic perceptual posteriors and the recovered posteriors. This procedure verified that the model parameters reside in an identifiable regime, thereby confirming that $Q(\theta)$ constitutes a mathematically sound ground-truth target.

Latent Behavioural State Discovery via GLM-HMM

To isolate periods of task engagement and account for trial-to-trial fluctuations in the animals' decision-making strategies, we modelled choice behaviour using a Generalised Linear Model Hidden Markov Model (GLM-HMM) [4]. This framework assumes that on any given trial t, the animal occupies one of K discrete latent states, $z_t \in \{1, \ldots, K\}$, with each state characterised by a distinct mapping (GLM weight vector, $\mathbf{w}_k$) from task variables to the binary choice probability.

The state-dependent choice probability was modelled as a Bernoulli distribution linked to a linear combination of five trial-history and stimulus predictors:

$$P(y_t = \text{Go} \mid \mathbf{x}_t, z_t = k) = f(\mathbf{w}_k^T \mathbf{x}_t) \tag{17}$$

where $f(\cdot)$ represents the link function. The predictor vector $\mathbf{x}_t$ comprised the current stimulus, the previous stimulus, the previous choice, the previous reward, and a constant bias term. To accurately reflect the varying sensory evidence, the current and previous stimulus orientations were normalised to the $[-1, 1]$ range (relative to the 45° category boundary) and scaled dynamically on a trial-by-trial basis by both the stimulus contrast (C_t) and a precision factor derived from the stimulus dispersion

(d_t). The precision factor was defined relative to the baseline training dispersion (5°), such that the effective stimulus regressor S_t was computed as:

$$S_t = \left(\frac{\theta_t - 45^\circ}{45^\circ}\right) \cdot C_t \cdot \left(\frac{5^\circ}{d_t}\right) \tag{18}$$

To ensure stable state discovery and maintain interpretability across the cohort, model parameters were optimised via Expectation-Maximisation (EM) using a rigorous three-stage hierarchical fitting procedure. First, a global template model was fit to the pooled data from all animals and sessions. We evaluated candidate models with $K \in \{2, 3, 4\}$ states, using 20 random initialisations per K, and selected the optimal number of latent states by minimising the Bayesian Information Criterion (BIC). To impose biological regularisation, an L2 penalty ($\lambda = 1.5$) was applied to the observation weights.

In the second stage, animal-specific models were fit using the transition matrix and observation weights of the optimal global model as a warm start. Finally, session-specific models were fit using the respective parent animal's parameters as priors. To prevent state-switching artifacts (label switching) across hierarchical levels, the resulting observation weight vectors of the animal and session models were systematically realigned to the global template using a Euclidean distance cost matrix solved via linear sum assignment.

Following parameter optimisation, we extracted the trial-by-trial posterior state probabilities and the Maximum A Posteriori (MAP) dominant state. This allowed us to explicitly partition our behavioural and neural analyses according to the animal's trial-to-trial engagement state.

Spatiotemporal Model Comparison of Choice Behaviour

To rigorously evaluate the predictive power of our Ideal Observer (IO) framework against empirical baselines, we conducted a comprehensive model comparison across the spatial extent of the virtual reality corridor. Because decision-making in our task is an extended spatiotemporal process, we independently evaluated all models across sequential spatial windows to track how the representations of sensory evidence and decision variables evolved as the animal approached the reward zone.

Spatial Windowing and Kinematic Feature Extraction

The virtual corridor preceding the reward zone was discretised into non-overlapping spatial windows of width $W = 10$ virtual units, evaluated at a step size of $S = 10$ units. For each trial t and spatial window w, we extracted the sum of pre-reward zone licks ($y_{lick,t,w}$) and the mean wheel velocity ($y_{vel,t,w}$). To control for variations in baseline motor vigour across days and subjects, these kinematic metrics were robustly Z-scored independently for each animal and spatial window.

Evaluated Models

Models were tasked with predicting the trial-by-trial binary choice (Go/No-Go) using 5-fold cross-validation. We evaluated two variants of our normative IO model against a nested family of Generalised Linear Models (GLMs).

Ideal Observer Models: While the IO model was originally parametrised to generate kinematics from the latent measurement m, here we inverted the model to predict the probability of a Go choice by conditioning the latent state on the observed kinematics. For a given trial, the likelihoods of the observed licks and velocity were combined with the generative prior $p(m \mid s)$ to compute a posterior distribution over the latent measurement:

$$p(m \mid s, y_{vel}, y_{lick}) = \frac{p(y_{vel} \mid m)p(y_{lick} \mid m)p(m \mid s)}{\int p(y_{vel} \mid m')p(y_{lick} \mid m')p(m' \mid s)dm'}$$

The predicted probability of a Go choice was then obtained by marginalising the measurement-conditioned choice probability (modelled via a softmax function over the decision variable, $DV(m)$)

over this inverted posterior:

$$P(\text{Go} \mid s, y_{vel}, y_{lick}) = \int \frac{1}{1 + \exp(-\beta_{decision} DV(m))} p(m \mid s, y_{vel}, y_{lick}) dm$$

We evaluated a *Full IO Model* conditioned on both licks and velocity, and a *Reduced IO Model* conditioned strictly on velocity.

Generalised Linear Models: As empirical baselines and performance ceilings, we trained a family of binomial GLMs with logit link functions, regularised via a Jeffreys prior (Firth's penalised likelihood) to mitigate quasi-complete separation issues in sparse spatial bins. The GLM family included:

- *GLM Ceiling*: Predictors included stimulus features (orientation, contrast, dispersion) alongside both kinematic variables (velocity, licks).
- *GLM Choice (Kinematics)*: Predicted choice from velocity and licks combined, velocity only, or licks only.
- *GLM Stimulus*: Predicted choice strictly from the stimulus features (s, c, d).
- *GLM History*: Predicted choice from the previous trial's action ($Choice_{t-1}$) and outcome ($Reward_{t-1}$).

Hierarchical Fitting Pipeline and Evaluation Metrics

To ensure the IO models were not disadvantaged by limited data in single folds, we maintained a two-stage hierarchical fitting pipeline evaluated independently for every spatial window. First, a global set of template parameters was obtained by fitting the pooled training data across all animals using Bayesian Adaptive Direct Search (BADS). Second, these global parameters seeded a multi-start optimisation procedure (comprising the group-level seed and multiple jittered initialisations) to fit the IO models to the individual animal's training data for that specific cross-validation fold. Model predictive performance on the held-out test folds was quantified using the Binomial Negative Log-Likelihood (NLL) and the Area Under the Receiver Operating Characteristic Curve (AUC). By computing these metrics iteratively across the discretised corridor, we generated spatiotemporal predictive profiles, allowing us to pinpoint the precise spatial locus at which the animals' kinematics dynamically coupled with their underlying perceptual and decision-related latent states.

Spatiotemporal Decoding of Stimulus Category

In parallel to predicting the animals' choices, we sought to determine how effectively the true underlying stimulus category (Go vs. No-Go) could be linearly decoded from the animals' evolving kinematics. This allowed us to quantify the temporal dynamics of sensory evidence accumulation and compare it against the normative predictions of the Ideal Observer (IO) model.

The spatial windowing, kinematic feature extraction, and hierarchical cross-validation pipelines were maintained exactly as described for the choice model comparison. However, the objective function was strictly altered to predict the true stimulus category (1 if the orientation fell within the Go category, i.e., $< 45°$, and 0 otherwise).

Evaluated Models for Stimulus Decoding

To prevent circularity, models containing explicit stimulus or trial-history predictors were strictly excluded from this analysis.

Ideal Observer Models: For the IO framework, we bypassed the decision-variable transformation. Crucially, to ensure a fair comparison against empirical baselines, the model was rendered blind to the true stimulus orientation on any given trial. First, we evaluated the probability mass of the perceptual posterior falling into the Go category for any hypothetical internal measurement m:

$$P(\text{Go Category} \mid m) = \int_0^{45°} p(s \mid m) ds \tag{19}$$

Next, we inferred the probability of the latent measurement m given strictly the observed kinematics and the marginal task prior $p(m) = \int p(m \mid s)p(s)ds$, completely independent of s_{true}:

$$p(m \mid y_{vel}, y_{lick}) = \frac{p(y_{vel} \mid m)p(y_{lick} \mid m)p(m)}{\int p(y_{vel} \mid m')p(y_{lick} \mid m')p(m')dm'} \tag{20}$$

The overall probability of the stimulus belonging to the Go category was then predicted by marginalising the perceptual category mass over this kinematics-restricted measurement posterior:

$$P(\text{Predicted Go} \mid y_{vel}, y_{lick}) = \int P(\text{Go Category} \mid m)p(m \mid y_{vel}, y_{lick})dm \tag{21}$$

As before, we evaluated a *Full IO Model* (conditioned on both velocity and licks) and a *Reduced IO Model* (conditioned strictly on velocity).

Generalised Linear Models: The empirical baselines were restricted to the purely kinematic GLM family. These models attempted to linearly discriminate the true stimulus category using logistic regression:

- *GLM Kinematic (Combined)*: Predicted stimulus category from velocity and licks.
- *GLM Kinematic (Velocity)*: Predicted stimulus category from velocity only.
- *GLM Kinematic (Licks)*: Predicted stimulus category from licks only.

Model performance across the spatial windows was similarly quantified using the Area Under the Receiver Operating Characteristic Curve (AUC).

Decoding Framework: Artificial Neural Network Architectures

To investigate the algorithmic representation of perceptual uncertainty in the primary visual cortex (V1), we trained artificial neural networks (ANNs) to decode the trial-by-trial perceptual posteriors, $Q(\theta)$, from V1 neural population activity. We contrasted two distinct decoding architectures, designed to mathematically instantiate the theoretical predictions of Probabilistic Population Codes (PPC) and Sampling-Based Codes (SBC).

Let $\mathbf{r}_{n,t} \in \mathbb{R}^U$ denote the Z-scored population activity vector of U neurons on trial n at time bin t, where $t \in \{1, \ldots, T\}$. Both decoding architectures rely on a shared multi-layer perceptron (MLP) backbone, denoted as $f_\phi(\cdot)$, comprising fully connected layers with ReLU activations. The architectures differ strictly in how they integrate information across time to produce the predicted posterior distribution over the stimulus orientation, $P(\theta)$.

Spatial Architecture (PPC)

Under the PPC framework, we hypothesise that the network represents uncertainty by integrating population activity spatially (across neurons) and temporally (across the stimulus epoch) *before* generating a continuous probability distribution. Therefore, the temporal integration occurs prior to the non-linear decoding step:

$$P_{PPC}(\theta \mid \mathbf{r}_n) = \text{Softmax}\left(f_\phi\left(\frac{1}{T}\sum_{t=1}^{T}\mathbf{r}_{n,t}\right)\right) \tag{22}$$

This architecture represents uncertainty explicitly in the geometry of the integrated population vector.

Temporal Architecture (SBC)

Conversely, the SBC framework posits that the network approximates distributions sequentially via independent temporal samples. Consequently, the ANN processes instantaneous population states, and the trial-level posterior is aggregated *after* the decoding step:

$$P_{SBC}(\theta \mid \mathbf{r}_{n,t}) = \text{Softmax}\left(f_\phi(\mathbf{r}_{n,t})\right) \tag{23}$$

$$\bar{P}_{SBC}(\theta \mid \mathbf{r}_n) = \frac{1}{T}\sum_{t=1}^{T} P_{SBC}(\theta \mid \mathbf{r}_{n,t}) \tag{24}$$

Loss Functions and Regularisation

The ANNs were trained to minimise the divergence between their predicted posteriors, $P(\theta)$, and the behaviourally-estimated perceptual posteriors, $Q(\theta)$, derived from the IO model. To ensure robustness, we optimised models across four distinct divergence metrics (denoted generally by the divergence operator $\mathcal{D}$).

Let θ_i represent the discrete orientation bins. The four evaluated divergence functions were defined as follows:

1. Kullback-Leibler (KL) Divergence:

$$D_{KL}(Q \parallel P) = \sum_i Q(\theta_i) \log\left(\frac{Q(\theta_i)}{P(\theta_i) + \epsilon}\right) \tag{25}$$

2. Jensen-Shannon (JS) Divergence:

$$D_{JS}(Q \parallel P) = \frac{1}{2}D_{KL}(Q \parallel M) + \frac{1}{2}D_{KL}(P \parallel M) \tag{26}$$

where $M(\theta_i) = \frac{1}{2}\left(Q(\theta_i) + P(\theta_i)\right)$.

3. 1D Wasserstein Distance: Computed as the L_1 distance between the discrete Cumulative Distribution Functions (CDFs) of the target and prediction:

$$W_1(Q, P) = \sum_i \left| \sum_{j \le i} Q(\theta_j) - \sum_{j \le i} P(\theta_j) \right| \tag{27}$$

4. PCA-Weighted Euclidean Distance: Distributions were projected onto the principal components (PCs) extracted from the training target sets. Let $\mathbf{v}_k$ denote the k-th PC eigenvector and λ_k its corresponding explained variance fraction. The loss was computed as the variance-weighted squared distance in the reduced subspace:

$$D_{PCA}(Q, P) = \sum_k \lambda_k \left(\mathbf{Q}^T\mathbf{v}_k - \mathbf{P}^T\mathbf{v}_k\right)^2 \tag{28}$$

Normative SBC theory posits that instantaneous population activity represents a single point estimate (or a highly localised Dirac delta function) drawn from the underlying posterior distribution, with uncertainty represented as variance *across* samples over time. Consequently, each independent instantaneous sample should possess minimal informational entropy. To enforce this theoretical constraint and prevent the SBC architecture from trivially learning a PPC-like representation at the single-bin level, we introduced an instantaneous entropy penalty, $\mathcal{H}$, scaled by a hyperparameter λ_H.

The total loss $\mathcal{L}$ for a given trial n under a chosen metric $\mathcal{D}$ was computed as:

$$\mathcal{L}_{SBC} = \mathcal{D}\left(Q(\theta), \bar{P}_{SBC}(\theta \mid \mathbf{r}_n)\right) + \lambda_H \frac{1}{T}\sum_{t=1}^{T} \mathcal{H}\left(P_{SBC}(\theta \mid \mathbf{r}_{n,t})\right) \tag{29}$$

where $\mathcal{H}(P) = -\sum_i P_i \log(P_i + \epsilon)$ computes the Shannon entropy of the instantaneous output distribution.

Because the PPC architecture is theoretically designed to explicitly parameterise a wide, continuous probability distribution at any given readout stage, it was exempt from the sharpness prior ($\lambda_H = 0$):

$$\mathcal{L}_{PPC} = \mathcal{D}\left(Q(\theta), P_{PPC}(\theta \mid \mathbf{r}_n)\right) \tag{30}$$

Temporal Integration Sweep and Validation

To determine the temporal scale at which SBC representations might degrade or collapse into PPC-like representations, we evaluated model performance across varying temporal bin sizes (50 ms to 250 ms). Activity traces were sliced (e.g., the full 2 s stimulus epoch, the late 1 s half-window, and the final 0.5 s quarter-window) and downsampled via a moving average prior to input.

Models were trained utilising a stratified train-test split to ensure uniform distribution of base stimulus orientations, contrasts, and dispersions between training and test sets. To establish a rigorous lower bound for baseline decoding failure, we trained parallel "shuffled" models where the trial-to-target mapping in the training set was randomised. All final divergence metrics were normalised to this shuffled baseline, where a ratio less than 1.0 indicates successful stimulus-driven inference.

Out-of-Distribution (OOD) Generalisation

To rigorously test whether the networks learned a generalisable structural manifold of uncertainty, rather than merely interpolating within the training distribution, we evaluated the architectures on Out-of-Distribution (OOD) generalisation splits. We engineered two orthogonal task manipulations: varying stimulus contrast at baseline dispersion, and varying stimulus dispersion at baseline contrast.

Networks were explicitly trained on one axis of manipulation and tested on the entirely unseen axis. For instance, in the *Contrast Generalisation* split, models were trained on trials featuring contrast variations (anchored by a 50% subset of baseline high-contrast/low-dispersion trials) and tested strictly on trials featuring dispersion variations (anchored by the remaining 50% of baseline trials). The *Dispersion Generalisation* split reversed this arrangement. This paradigm allowed us to penalise models that overfit to specific low-level sensory statistics, explicitly testing whether the temporal (SBC) or spatial (PPC) integration schemes natively abstracted the concept of uncertainty.

References

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